Fullstack Development

API Architectures and Design #1

Content

- **What is API?**
- **API Architecture Styles**
- RESTful API design
- API Security
- API Testing

What is API?

Application Programming Interfaces

- **Web service**
- Provides communication and integration between software systems
- Data exchange between components,
- Remote function calls

API Architecture Styles

Top 6 most popular styles:

SOAP

- **Simple Object Access Protocol**
- XML-based for enterprise application, mature and comprehensive
- Commonly used in financial services

RESTful

- **REpresentational State Transfer**
- Build on top of `HTTP methods`, popular and easy to implement
- Resource-based, rely on multiple endpoints, return fixed data structures
- Not suitable for `real-time` applications or highly connected model

RESTful Example

RESTful: E-Commerce

Method	Endpoint	Description
GET	/products	fetch a list of products
GET	/users/{userId}	fetch user details
GET	/orders/{orderId}	fetch order details
...		

For `user order history`, multiple requests are required

RESTful: Social Media Platform

Method	Endpoint	Description
GET	/users/{userId}/posts	fetch a user's posts
GET	/users/{userId}/followers	fetch a user's followers
GET	/posts/{postId}/comments	fetch comments on a posts
...		

Each endpoint returns a fixed set of data, requiring multiple requests to gather all necessary information

RESTful

Use RESTful API when:

- API is simple and `CRUD-based`
- Want to use `HTTP caching`
- Building `public APIs` or microservices
- Prefer `minimal tooling` or need API to support older systems

GraphQL

- `Architecture` and `query language`
- Allows clients to specify exact data they need (no `over-/under-fetching` data)
- Strong typed API (predefined `schema`)
- Operates through a single endpoint (flexible and efficient)
- **Hard** to optimized `database query` and `cache`

GraphQL Example

GraphQL: E-Commerce

```graphql
query GetUserOrders {
  user(id: "userId") {
    id
    name
    gender
    orders {
      id
      total
      items {
        product
        quantity
        price
      }
    }
  }
}
```

GraphQL: Social Media

```
query GetUserPosts {
  user(id: "userId") {
    id
    name
    posts {
      id
      content
      comments {
        id
        author {
          id
          name
        }
        content
      }
    }
  }
}
```

GraphQL

Use GraphQL when:

- Frontend needs custom or `nested data structure`
- Want to reduce the `number of API calls`
- Building and `SPA` or `mobile app` with dynamic UI
- Need `flexibility` and scalability

GraphQL vs. RESTful Comparison

gRPC

- Open source **Remote Procedure Call** framework created by Google
- High performance for microservices
- Using `Protocol Buffer` (aka. `Protobuf`) which is **binary encoded**

gRPC

Why is gRPC very fast?

- Send data in binary encoded format (`Protobuf`)
- Run on HTTP/2 protocol

gRPC

gRPC - How does it work?

WebSockets

- `Bi-directional` for `low-latency` data exchange over TCP connection
- `Real-time` and `persistent connection` (for Live chat, real-time gaming)

WebSockets

JavaScript/TypeScript library

ws - A simple WebSocket

```
npm install ws
npm install --save @types/ws
```

Socket.IO

```
npm install socket.io
npm install --save-dev typescript ts-node @types/socket.io
```

Webhook

- `Asynchronous` for `even-driven` application
- `Webhook consumer` must register with `webhook provider`
  - When certain event occurs, `webhook provider` invokes `HTTP request` to `webhook consumer` at a `certain URL`
  - The `webhook consumer` then handles the request in certain way (e.g., notify user)

Webhook

Stripe

- Payment service
- Allow developer to register multipel webhooks, select specific `payment-related` events

GitHub

- Able to notify developer about `code commits`, `pull requests`, and `issues`

Twilio

- Communication service
- Able to notify developers about `SMS` and `voice-related` events

References

- GraphQL Example: Apollo Server | Apollo Client
- REST vs. GraphQL: Choosing the right API for your project
- GraphQL vs. REST: Top 4 advantages & disadvantages
- How to integrate gRPC with React and TypeScript
- Using gRPC in React the Modern Way
- Chat App driven by WebSockets using Socket.IO and TypeScript